

5. [Maximum mark: 6]

The random variables X and Y are normally distributed with $X \sim N(7, a^2)$ and $Y \sim N(19, a^2)$, where $a > 0$.

- (a) Find b such that $P(X > b) = P(Y > 22)$. [2]
- (b) Write down the approximate value of $P(7 - a < X < 7 + a)$, correct to two significant figures. [1]
- (c) Given that $a = 3$, calculate the approximate value of $P(Y < 22)$, correct to two significant figures. [3]

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